



Solution Set

Question 1

Consider a European option priced from a characteristic function. Define the state variables (log price, characteristic function, damping parameter, quadrature grid, and FFT spacing) and derive the Fourier inversion relation. State the measure, filtration, and tradability assumptions.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

Define the tradable state variables first: log price, characteristic function, damping parameter, quadrature grid, and FFT spacing. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for Fourier inversion relation. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$\begin{aligned} \varphi_X(u) &= E[e^{iuX}], & f_X(x) &= \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du \\ d\Pi &= dV - \Delta dS, & d\Pi &= r\Pi dt \end{aligned}$$

Question 2

For a European option priced from a characteristic function, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$\begin{aligned} \varphi_X(u) &= E[e^{iuX}], & f_X(x) &= \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du \\ Z_t &= \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, & E^Q[X | \mathcal{F}_t] &= \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t] \end{aligned}$$

Question 3

Identify the discounted tradable or density-adjusted process in a European option priced from a characteristic function. Prove the martingale condition and name one modelling choice that would break it.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is aliasing from an inconsistent FFT grid.

$$\begin{aligned} \varphi_X(u) &= E[e^{iuX}], & f_X(x) &= \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du \\ E[M_t | \mathcal{F}_s] &= M_s, & 0 &\leq s < t \end{aligned}$$

Question 4

Using a lognormal characteristic function evaluated on a grid from -20 to 20, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

Use the stated market inputs: a lognormal characteristic function evaluated on a grid from -20 to 20. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$V_t = e^{-r\Delta t} (qV_u + (1-q)V_d)$$

Question 5

For a European option priced from a characteristic function, compute and interpret bump-and-reprice Greeks from the Fourier price. Explain the hedge sign and the unit of each sensitivity.

Answer key

The relevant exposure is bump-and-reprice Greeks from the Fourier price. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \nu d\sigma$$

Question 6

Suppose the desk observes Heston or lognormal characteristic-function parameters fit to vanilla prices. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Answer key

The calibration instruments are Heston or lognormal characteristic-function parameters fit to vanilla prices. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$\hat{\theta} = \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2$$

Question 7

State the terminal and boundary conditions for a European option priced from a characteristic function. Explain how integrability, damping, and payoff transform conditions prevents economically impossible prices.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is integrability, damping, and payoff transform conditions.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 8

Design a Monte Carlo or lattice estimator for a European option priced from a characteristic function. Specify numerical inversion checked against Black-Scholes, the estimator, the discounting rule, and the standard error.

Answer key

Simulate numerical inversion checked against Black-Scholes. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 9

A junior quant implements a European option priced from a characteristic function and gets a stable but wrong answer. Explain how aliasing from an inconsistent FFT grid creates the error and design a diagnostic test.

Answer key

The main implementation risk is aliasing from an inconsistent FFT grid. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$\text{error}(N, \Delta t) = \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 10

Write an NumPy algorithm for a European option priced from a characteristic function. Include inputs, outputs, diagnostics, and one benchmark case.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$\begin{aligned} \varphi_X(u) &= E[e^{iuX}], & f_X(x) &= \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du \\ \text{tree cost} &= O(n^2), & \text{simulation cost} &= O(NM) \end{aligned}$$

Question 11

Starting from the payoff definition of a European option priced from a characteristic function, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Answer key

Define the tradable state variables first: log price, characteristic function, damping parameter, quadrature grid, and FFT spacing. The payoff is then written at the relevant horizon and valued under the pricing measure.

The derivation uses the economic relation for Fourier inversion relation. If a hedge or replication argument is used, the stochastic exposure is neutralized locally and the remaining cash account earns the risk-free or collateral rate.

The final expression must agree with the payoff at maturity and with the obvious limiting cases.

$$\begin{aligned} \varphi_X(u) &= E[e^{iuX}], & f_X(x) &= \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du \\ d\Pi &= dV - \Delta dS, & d\Pi &= r\Pi dt \end{aligned}$$

Question 12

Create a stressed-market version of a lognormal characteristic function evaluated on a grid from -20 to 20. Recompute the value or risk metric and explain which term dominates the change.

Answer key

Use the stated market inputs: a lognormal characteristic function evaluated on a grid from -20 to 20. Compute future state values first, then take the correct conditional average, then discount.

Keep a calculation table with states, probabilities, payoffs, discount factors, and final present values. This makes unit mistakes visible.

The answer should include a sign and magnitude check, not only a final number.

$$\begin{aligned} \varphi_X(u) &= E[e^{iuX}], & f_X(x) &= \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du \\ V_t &= e^{-r\Delta t} (qV_u + (1-q)V_d) \end{aligned}$$

Question 13

Use a second-order Taylor approximation for a European option priced from a characteristic function. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

The relevant exposure is bump-and-reprice Greeks from the Fourier price. Compute the first-order term, then add convexity or cross terms only when the state variables require them.

A hedge has the opposite sign of the exposure. Delta is per unit of underlying, DV01 is per basis point, and vega is per volatility point or percent depending on convention.

After hedging, report the residual risk that remains.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \nu d\sigma$$

Question 14

Show how a change of numeraire or change of measure would simplify one expectation in a European option priced from a characteristic function. Identify what becomes a martingale after the change.

Answer key

The pricing measure is chosen so discounted tradable prices are martingales under the chosen numeraire.

The Radon-Nikodym density reweights probabilities; it does not change the contractual payoff. In Brownian models, Girsanov changes the drift and leaves the quadratic variation unchanged.

A useful check is that the density has expectation one and that the resulting discounted price process has zero drift.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$Z_t = \frac{dQ}{dP} \Big|_{\mathcal{F}_t}, \quad E^Q[X | \mathcal{F}_t] = \frac{1}{Z_t} E^P[Z_T X | \mathcal{F}_t]$$

Question 15

Give a proof-style answer showing when the wealth process for a European option priced from a characteristic function is self-financing. Explain why predictable trading strategies matter.

Answer key

The martingale candidate is the discounted tradable price, or the density-adjusted process after a measure change.

Condition on the earlier information set and verify that the conditional expectation equals the earlier value. Any remaining predictable drift indicates a missing funding, dividend, numeraire, or model adjustment.

For this topic, the practical issue is aliasing from an inconsistent FFT grid.

$$\varphi_X(u) = E[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$E[M_t | \mathcal{F}_s] = M_s, \quad 0 \leq s < t$$

Question 16

Market quotes for Heston or lognormal characteristic-function parameters fit to vanilla prices are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

The calibration instruments are Heston or lognormal characteristic-function parameters fit to vanilla prices. Write a weighted least-squares objective using market quotes and model quotes.

Weights should reflect liquidity, bid-ask width, and quote reliability. Parameter constraints should prevent impossible prices or unstable extrapolation.

After calibration, inspect residuals and check whether each parameter has a distinct effect on the quotes.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$\hat{\theta} = \arg \min_{\theta} \sum_i w_i (M_i(\theta) - Q_i^{\text{mkt}})^2$$

Question 17

For a European option priced from a characteristic function, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Answer key

The terminal condition comes from the payoff. Boundary conditions come from economic limits such as zero spot, very large spot, barrier touch, exercise, or discount-factor positivity.

For this product, the key boundary idea is integrability, damping, and payoff transform conditions.

A boundary condition is wrong if it permits arbitrage, negative values for non-negative payoffs, or a price that violates obvious monotonicity.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$V(T, S) = g(S), \quad V(t, S_{\text{boundary}}) = \text{economic boundary value}$$

Question 18

Design a convergence study for numerical inversion checked against Black-Scholes. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Answer key

Simulate numerical inversion checked against Black-Scholes. For each path, compute the payoff using only information available along that path.

The estimator is the discounted sample average. Report standard error and rerun with larger path counts or finer time steps to separate sampling error from discretization error.

For path-dependent products, store the running state variable explicitly.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N X^{(i)}, \quad \widehat{\text{SE}} = e^{-rT} \frac{s_X}{\sqrt{N}}$$

Question 19

Find one way that aliasing from an inconsistent FFT grid could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$

Answer key

The main implementation risk is aliasing from an inconsistent FFT grid. This can produce stable but wrong results because the error is systematic.

A good diagnostic compares the implementation with a limiting case or benchmark where the direction of the answer is known.

Change one input at a time and verify monotonicity, units, and convergence.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$
$$\text{error}(N, \Delta t) = \widehat{V}_{N, \Delta t} - V_{\text{benchmark}}$$

Question 20

Explain a European option priced from a characteristic function and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.

Answer key

Create clear arrays for inputs, states, probabilities, discount factors, and payoffs. Check dimensions before combining them.

Return both the main value and diagnostics such as convergence, residuals, standard error, or benchmark difference.

The routine is acceptable only if the result can be reproduced from the displayed assumptions and data.

$$\varphi_X(u) = \mathbb{E}[e^{iuX}], \quad f_X(x) = \frac{1}{2\pi} \int e^{-iux} \varphi_X(u) du$$
$$\text{tree cost} = O(n^2), \quad \text{simulation cost} = O(NM)$$